BISHAL RANJITKAR

AI Engineer | Generative AI | AI Automation | LLM Applications | RAG & Agentic Systems
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PROFESSIONAL SUMMARY

AI Engineer focused on building and deploying practical Generative AI systems, LLM applications, RAG, AI agents, and automation solutions. Experienced with Python, FastAPI, AWS, Docker, APIs, and AI system deployment. Skilled in turning business problems into reliable and scalable AI solutions. Interested in building production-ready AI systems that improve efficiency and reduce manual work. AWS Certified Solutions Architect Associate.

TECHNICAL SKILLS

- **AI & Generative AI:** Generative AI, LLM Applications, RAG, AI Agents, Prompt Engineering, LLM Evaluation, Multi-Agent Systems, Embeddings, Vector Databases
- **AI Engineering:** Python, FastAPI, REST APIs, AI System Design, Workflow Automation, API Integration
- **AI Frameworks & Orchestration:** LangChain, LangGraph, n8n, Model Context Protocol (MCP), LangSmith
- **Cloud & Infrastructure:** AWS, Docker, Terraform, CI/CD, Linux
- **Databases:** PostgreSQL, MongoDB
- **Development & Tools:** Git, GitHub, Postman, LangSmith

PROFESSIONAL EXPERIENCE

Associate AI Automation Engineer
Sundhara, Kathmandu, Nepal
Recruit Nepal PVT.LTD
Jan 2026 – Present

- Optimize production AI systems for cost, latency, reliability, and token efficiency while maintaining response quality across multi-agent recruitment workflows.
- Extend and refactor enterprise RAG and multi-agent architectures, adding new capabilities while improving existing production pipelines.
- Engineer reliability improvements including error recovery, monitoring, performance optimization, and workflow resilience under limited compute and budget constraints.
- Implement AI engineering practices including prompt engineering, agent-based tooling, and structured workflow design to improve system reliability and scalability.
- Strengthen AI workflow security through input validation, access controls, and guardrails against prompt injection and unauthorized data exposure.

AI Automation Intern
Sundhara, Kathmandu, Nepal
Recruit Nepal PVT.LTD
Oct 2025 – Dec 2025

- Architected a multi-channel AI automation ecosystem that reduced manual recruitment and marketing operations by 90%.
- Engineered an AI auto-reply system using specialized AI agents, RAG, embeddings, and vector databases for automated candidate communication.
- Developed a lead-generation pipeline that scraped and enriched contact data from 500+ international technology companies, reducing outreach time by 85%.
- Implemented production reliability features including real-time error-handling webhooks and automated Slack monitoring alerts.
- Integrated multimodal AI for automated branding and social media content generation across LinkedIn, Facebook, and Instagram.

Cloud Apprentice
Ekantakuna, Lalitpur, Nepal
ADEX INTERNATIONAL PVT.LTD
July 2025 – August 2025

- Completed an 8-week hands-on AWS Cloud Apprenticeship.
- Worked on real-world AWS services (EC2, S3, RDS, IAM, VPC).
- Prepared for the AWS Solutions Architect Associate exam through labs and projects.

PRATICAL PROJECTS

- **LangGraph RAG Chatbot: LangGraph | RAG | FAISS | Embeddings | MCP | SQLite | LangSmith | Streamlit**
  - Deployed a 2-mode LangGraph chatbot on Streamlit Cloud, with streaming replies.
  - Added 4 built-in tools and 3 MCP servers, each of which the user can turn on or off.
  - Built Constitution RAG with 700-character chunks, MiniLM embeddings, FAISS, and the top 6 passages.
  - Saved each chat as its own thread in 1 SQLite database, so old chats reopen after a restart and traced each chat in LangSmith with the same thread id.

- **YouTube Chatbot using RAG: Python | LangChain | LLM | RAG | Embedding Model**
  - Built an LLM-powered chatbot that enables users to interact with and ask questions about YouTube video content.
  - Implemented a retrieval-based architecture to process video information and provide context-aware responses.
  - Applied LangChain components to connect document processing, retrieval, prompting, and LLM response generation.
  - Added conversational question-answering capabilities so users can query video content and receive relevant answers.

- **Automated Multimodal Branding & Social Media Sync: : n8n, API Integration, MCP, Webhook:**
  - Developed an automated branding workflow using n8n, webhooks, Gemini AI, and API integrations.
  - Built a system to detect newly published website vacancies through webhook triggers.
  - Extracted key job details and generated branded captions, post copy, and design prompts automatically.
  - Used multimodal AI logic to support vacancy-based creative content generation for social media.
  - Synchronized job post content across LinkedIn, Facebook, and Instagram for consistent brand visibility.
  - Reduced manual social media posting work and improved recruitment marketing efficiency.

TRAINING & CERTIFICATIONS

- **AWS Solutions Architect Associate:** Gained in-depth knowledge of AWS core services such as EC2, S3, RDS, IAM, and VPC. Learned to design and deploy scalable, highly available, and fault-tolerant systems on AWS.
- **Retrieval-Augmented Generation:** Mastered Retrieval-Augmented Generation principles, including document chunking, vector embeddings, semantic search retrieval, and contextual prompt injection for LLM applications.
- **Model Context Protocol:** Gained hands-on expertise building and integrating Model Context Protocol (MCP) servers and tools to connect AI models seamlessly with custom external data sources and APIs.
- **n8n Level 2 Certified:** Built on foundational knowledge to implement advanced workflow automation, including complex data merging, batch processing, API integrations, error handling, and automated reporting with scalable workflows.
- **Automate Tasks with Python:** Learned to automate repetitive tasks using Python, including file handling, data processing, and basic scripting to improve efficiency and reduce manual effort.
- **Backend Development:** Proficient in Node.js, Express, MongoDB and Postgres SQL for building scalable server-side applications and Python for task automation.

EDUCATION

TRIBHUVAN UNIVERSITY / 2020 – 2025
Bachelor of Computer Applications (BCA) / GPA: 3 .29 /4.0
Relevant Subjects: Python Programming,Database Management System, Software Engineering, Operating Systems, Web Technology, Computer Networking, Cloud Computing, Information Security

EVEREST INNOVATIVE COLLEGE / 2018 – 2020
+2 Science, National Examination Board (NEB) / GPA: 3.4/4.0
Relevant Subjects: Physics, Mathematics, Chemistry, Computer Science, English